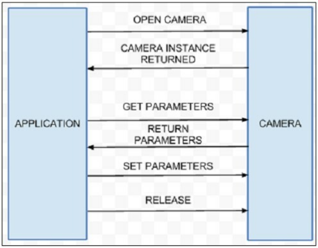

Figure 1: Design

```
        android:layout_height="fill_parent">
        <TextView android:layout_width="fill_parent"
          android:layout_height="wrap_content"
        android:text="@string/app_title" android:gravity="center"
          android:padding="20dip" android:textSize="25sp"/>
        <ToggleButton android:layout_height="wrap_content"
          android:layout_width="match_parent"
android:text="ToggleButton"
          android:id="@+id/toggleButton1"
        android:padding="20dip"></ToggleButton>
</LinearLayout>
```

Now, for the business logic. You should get the *Flashlight* parameters, then set the flash mode of the *Flashlight* to *"FLASH_MODE_TORCH"* to switch on, and reset the *Flashlight* to *FLASH_MODE_OFF* to switch off. Here's the code:

```
public class MainActivity extends Activity {
        Camera cam;
        ToggleButton mTorch;
        Parameters camParams;
        private Context context;
        AlertDialog.Builder builder;
        AlertDialog alertDialog;
        private final int FLASH_NOT_SUPPORTED = 0;
        private final int FLASH_TORCH_NOT_SUPPORTED = 1;
        /** Called when the activity is first created. */
        @Override
        public void onCreate(Bundle savedInstanceState) {
          super.onCreate(savedInstanceState);
          setContentView(R.layout.main);
          context = MainActivity.this;
          if(context.getPackageManager().hasSystemFeature(
              PackageManager.FEATURE_CAMERA_FLASH)){
              mTorch = (ToggleButton) findViewById(R.
id.toggleButton1);
```

Figure 2: Alert Dialogue

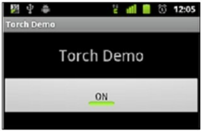

Figure 3: Torch Demo (ON)

```
              ....... logic......
        else{
              showDialog(MainActivity.this, FLASH_NOT_SUPPORTED);
          }
```

Now the first condition to check is whether the device has the flash feature; if yes, then proceed, else show an alert dialogue like in Figure 2. Then you get a reference to your *ToggleButton* using the *findViewById()* method.

```
        mTorch.setOnCheckedChangeListener(new
OnCheckedChangeListener() {

              @Override
              public void onCheckedChanged(CompoundButton
buttonView,                          boolean isChecked) {

                  try{
                      if(cam != null){
                          cam = Camera.open();
                      }
```

You then need to override the *ToggleButton's onCheckedChanged* method to handle state toggles. The Boolean parameter *isChecked* specifies whether the state is ON or OFF. Next, get hold of the camera object to access it:

```
                      camParams = cam.getParameters();
                      List<String> flashModes =
              camParams.getSupportedFlashModes();
                      if(isChecked){
                          if(flashModes.contains(
                              Parameters.FLASH_MODE_TORCH)) {
              camParams.setFlashMode(
                              Parameters.FLASH_MODE_TORCH);
                      }else{
                          showDialog(MainActivity.this,
                          FLASH_TORCH_NOT_SUPPORTED);
                          }
                      }else{
                      camParams.setFlashMode(Parameters.FLASH_MODE_
OFF);
                          }
```